Do Now · retrieval

Lesson 046 · Objects in Arrays and Lists

Name: _____ Class: _____ Date: _____

Answer four retrieval questions before reviewing earlier learning.

Q1

What is a constructor used for?

Q2

What does private mean for an instance field?

Q3

What index identifies the first C# array item?

Q4

A rising-edge D-type flip-flop starts with Q=0. D=1 at the first rising edge. D then falls to 0 before the falling edge and stays 0. Trace Q immediately after the first rising edge, the falling edge and the next rising edge.

New Learning · key ideas

The important knowledge from today's lesson. Read, annotate, and explain it.

Core idea 1

Observe — A class describes an object; the array stores references to two Book instances.

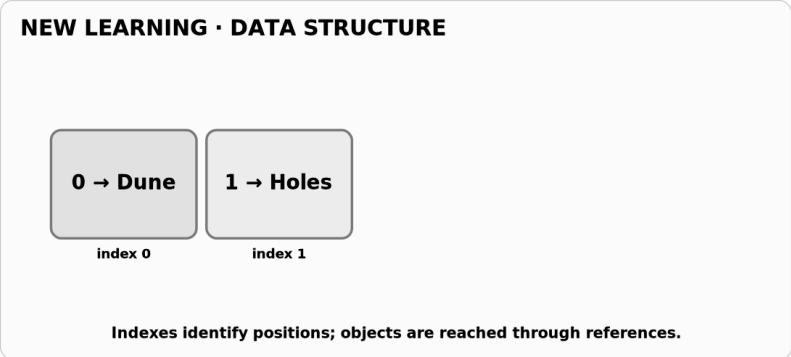
Core idea 2

Explain — Indexes start at 0. shelf[1] identifies the second reference, then .Title reads its property.

Core idea 3

Apply — An array has a fixed length after creation; individual array elements can refer to objects whose properties change.

Key visual · complete concept



Annotate the key visual

Explain which labels or features show the main process.

New Learning · applying the ideas

Vocabulary · use the correct technical terms

object — Instance of a class.

reference — Value identifying an object instance.

aliasing — Several references to the same object.

index — Position in a sequence.

Check your understanding

Explain one key process in your own words using two of the terms above.

More animated examples are available in the original online lesson.

I Do · worked example

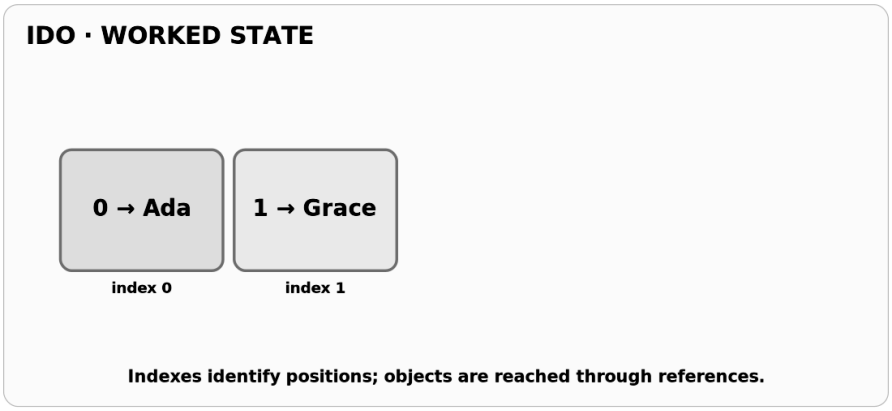
Problem

Trace the title stored at books[0] in a two-object Book array.

Follow the worked method

1. Trace and justify — Index 0 refers to the first object; its Title is Ada.

Source model · final illustrated step



Why this method works

Identify the decision, rule or invariant that makes the method correct.

We Do · Try → Discuss → Improve

TRY independently in BLACK PEN; compare methods with a partner; improve your reasoning.

Guided problem · TRY

Trace and justify

Index 1 refers to the second object. The property update changes its Title; output is Updated.

DISCUSS

Compare how you reached the answer. Where could someone make an error?

IMPROVE

Add one precise correction or a stronger reason. Answers are not printed here.

You Do · independent practice

Answer these selected original questions in BLACK PEN. Other exam tasks remain online.

Question 1

Given shelf titles Dune,Holes, give output of shelf[1].Title and reason. [4]

Question 2

Write C# to create an array of two Book objects with titles A and B, then print first title. [6]

Check that you have used the technical terms and explained your decisions.

You Do · extended practice

Answer these selected original questions in BLACK PEN. Other exam tasks remain online.

Question 3

Two array elements reference the SAME Book object. Changing its title via first reference is then printed via second. Explain. [6]

Check that you have used the technical terms and explained your decisions.

Exit Ticket · reflect and retrieve

Exit Ticket

Updating a property through one reference can be visible through another alias.

Use key vocabulary from memory

1. Explain object in your own words.

2. Explain reference in your own words.

3. Explain aliasing in your own words.

Confidence and next step

How confident are you? Thumbs up / side / down

Which question would you improve after checking the classroom model?

Check answers after the lesson using the teacher-guided source models.